

# WorkInSync Access Card Management Integration - API based

| Document Control |                                                           |
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| Ver No.         | Author        | Approved by    | Approved Date | Description                                 |
| 1.0             | Rahul Agrawal | Ujjwal Trivedi | 28-Jun-2022   | Initial Doc                                 |
| 1.1             | Binoy Dedhia  | Ujjwal Trivedi | 29-Nov-2022   | Updated version (with actual API signature) |
| 1.2             | Aditya Dutta  | Ujjwal Trivedi | 19-Jul-2024   | Addition of RFID                            |

## Purpose

This document explains the process and data required for integrating WorkInSync system with client's own/third party access management vendors using APIs.

## Prerequisite

A Client specific username and password will be shared with the client.

## Description

Data Integration is to facilitate regular syncing of data between Client/vendor systems and WorkInSync via rest-based APIs. The following services from WorkInSync would be involved:

**API Gateway:** The API gateway will facilitate all interactions between the client service and the WorkInSync services. Current assumption is that all APIs will be client/vendor initiated interactions. Key aspects of this gateway like throttling, fault tolerance, secure data exchange mechanism, error logging and communication need to be discussed in further detail.

## Authentication

The Workinsync API interface uses a Bearer token mechanism for APIs provided to the clients. A Client specific username and password will be shared with the client. Before calling any other APIs, GetAuth token API needs to be called to receive an access token. On success, a short lived access token will be received in response to this API call. This access token in turn needs to be passed in the header of each API call as a 'bearer' token. If at any point the current authorization token expires, the 'Get AuthToken' API needs to be called again to receive a fresh access token.

**Endpoint:** POST {baseUrl}/auth/token

baseUrl: <https://api.moveinsync.com>

### Headers:

- Authorization: Basic <base64(client\_id:client\_secret)>

client\_id = username

client\_secret = password

**Body (form-encoded):**

- grant\_type=client\_credentials

**Sample cURL:**

```
curl --location 'https://api.moveinsync.com/auth/token' \
--header 'Authorization: Basic
cHdjYWMTUFdDQUM6Mzg2MWNlZmQtN2M5Ni00YzlkLTkyNTctYWRhZjUyZDM2NTQ4' \
--form 'grant_type="client_credentials"'
```

**Sample response:**

Success:

```
{
  "access_token": "eyJhbGciOiJSUzI1Ni.....DyCIIJYEMA4g",
  "token_type": "Bearer",
  "expires_in": 172799
}
```

## API Details

**Endpoint:** POST {baseUrl}/integration/bookings/ci-co

**baseUrl:-** <https://api.moveinsync.com>

**Header:**

- Authorization: Bearer <Auth token> → generated from above authentication API

### Request format from Access control system

| Request format |        |                                                                           |         |                                             |
|----------------|--------|---------------------------------------------------------------------------|---------|---------------------------------------------|
| Field Name     | Format | Field Mapping                                                             | Length  | Constraints                                 |
| filter*        | String | EmployeeID or<br>EmployeeName<br>or<br>EmployeeEmailID<br>of the Employee | Max: 50 | Should be unique<br>within each<br>employee |

|                |        |                                                                                                                            |          |                                                                                                               |
|----------------|--------|----------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------|
|                |        | scanning so that we can check the booking for them                                                                         |          |                                                                                                               |
| officeName     | String | Office name for where the action was performed                                                                             | Max: 100 |                                                                                                               |
| bookingStatus* | String | Defines the action to be taken i.e. checkin in this case                                                                   | Max: 50  | Send "SIGNED_IN" so that we know we have to check in or "SIGNED_OUT" so that we know we have to checkout      |
| epochTime*     | Long   | Time Stamp at which the above activity has been performed                                                                  |          | Date format will be in Epoch format in milliseconds                                                           |
| premiseId      | String | The unique ID associated with the location where the action was performed. It may be an office or specific floor location. | Max: 50  | Reader ID to office/ floor ID mapping would be required as an one time activity                               |
| readerId       | String | The unique ID associated with the device at the entry point.                                                               | Max: 50  | Reader ID to office/ floor ID mapping would be required as an one time activity                               |
| rfid           | string | Unique identification for a card No.                                                                                       | Max: 50  | Would need to create a mapping for RFID with corresponding employees. Filter can be skipped if this is passed |

## Response format

| Response format |         |                                |          |                                |
|-----------------|---------|--------------------------------|----------|--------------------------------|
| Field Name      | Format  | Field Mapping                  | Length   | Description                    |
| status          | Integer | API status code                | Max : 4  | Represents API success/failure |
| data            | UUID    | Booking Id                     | Max : 50 |                                |
| message         | String  | Success/Failure message of API |          |                                |

## Actions taken based on access control information received

- Check-in/Check-out to an existing booking of an employee
- If the createBookingIfNotPresent flag is true, we will create an office booking for employees who do not have an office booking based on scan time and end time.

## Sample API request and response

### Prerequisites

- We're assuming that the scanner device will communicate with WorkInSync system in real time
- We will need clients to provide mentioned information about the vendors that will consume the REST APIs via API gateway. It is required for registering clients for authentication and authorization on the gateway
- Without proper Auth, external 3rd parties can't access our internal APIs

### Note:

- This is a sample API and it may change

## Sample request payload

### Sample payload 1:

```
{
  "bookingStatus": "SIGNED_IN",
  "epochTime": 1668428940000,
  "filter": "qa134@mis.com",
  "premiseId": "",
  "readerId": "wertyhbf!239df",
  "officeName": "",
  "rfid": ""
}
```

### Sample payload 2:

```
{
  "bookingStatus": "SIGNED_IN",
  "epochTime": 1668428940000,
  "filter": "",
  "premiseId":
    "7107f124-cc4d-4495-8121-81e7f6594252/864d2c3-a1e8-4e2c-a646-cae76b36d5a2",
  "readerId": "",
  "officeName": "Prestige Trade Center",
  "rfid": "2959067"
}
```

### Instructions regarding the payload:

1. BookingStatus can be SIGNED\_IN/SIGNED\_OUT. SIGNED\_IN status will check-in into a booking or create a booking and check-in into it based on the property. SIGNED\_OUT will sign out the employee from the existing booking.
2. epochTime is the time of check-in/check-out.
3. filter accepts either emailId/phone number/empID. If there is rfid associated with the employee, then it can be passed in the rfid variable by leaving the filter variable as blank.
4. ReaderId or premiseId or officeName need to be passed for identifying the office.

## Sample response

### Success

i:

API response status code will be 200

```
{
  "status": 200,
  "data": "832dbe42-f3eb-48cd-91d8-8c81ccf24ca7",
  "message": "Check-in successful"
}
```

ii:

API response status code will be 200

```
{
  "status": 200,
  "data": "832dbe42-f3eb-48cd-91d8-8c81ccf24ca7",
  "message": "Check-out Successful"
}
```

### Failure

i:

If there is any internal failure:

API response status code will be 200

```
{
  "status": 1001,
  "data": null,
  "message": "Some problem processing your request"
}
```

ii:

If booking is not found:

API response status code will be 200

```
{
  "status": 1001,
  "data": null,
  "message": "Booking is not present for rfid: 123"
}
```

